Max Slope (MS) Step Distribution Statistics for KL divergence

Activation Type	Absolute Step Size			Relative Step Size		
	Mean	Std dev	KS	Mean	Std dev	KS
Model Generated	45.51	12.89	0.00	56.34	9.08	0.00
Random	58.54	**12.33**	0.47	71.83	**11.75**	0.69
Synthetic Baseline	54.79	14.02	0.32	62.41	12.05	0.32
Synthetic Structured	**48.79**	15.64	**0.13**	**61.86**	13.51	**0.26**

Table C.1: We find that our results for KL divergence of next-token prediction probabilities are similar to those for L2 distance at Layer 11. This table contains the mean, standard deviation and KS statistic for MS step distributions for all the perturbations we perform. The KS statistic is measured against perturbations towards model-generated activations, with a lower value meaning higher similarity.

D Isolating the effect of SAE reconstruction error

We denote the reconstruction of an activation A with `SAE(A) = decode(encode(A))`. To isolate the effect of SAE reconstruction error on the blowup location, we examine perturbations towards a reconstruction of a model-generated target activation `SAE(T)`. We compare these to perturbations towards model-generated activations and find that they are very similar, with blowups occurring slightly later for perturbations towards SAE reconstructions (Figure D.1, Table D.1). We also find that reconstructions of model-generated activations also have plateaus around them. This shows that the majority of the difference in our synthetic activations comes from the heuristics we use to select latents, and not the SAE reconstruction error.

This similarity suggests that SAE reconstructions behave like model-generated activations for the most part, and that the reconstruction error causes a small systematic shift in the blowup location. This points to some information loss that causes the model to respond slightly less to perturbations towards SAE reconstruction, which is relevant for interpreting experiments that use SAE latents.

Max Slope (MS) Step Distribution Statistics for SAE reconstructions

Activation Type	Absolute Step Size			Relative Step Size		
	Mean	Std dev	KS	Mean	Std dev	KS
Model Generated	41.11	10.40	0.00	51.60	7.82	0.00
Random	52.49	**10.21**	0.45	65.01	11.19	0.61
SAE Reconstruction	**41.49**	11.34	**0.02**	**53.34**	**8.39**	**0.11**

Table D.1: We find that perturbations towards model-generated activations are almost identical to perturbations towards their SAE reconstructions. This table contains the mean, standard deviation and KS statistic for MS step distributions for all the perturbations we perform. The KS statistic is measured against perturbations towards model-generated activations, with a lower value meaning higher similarity.

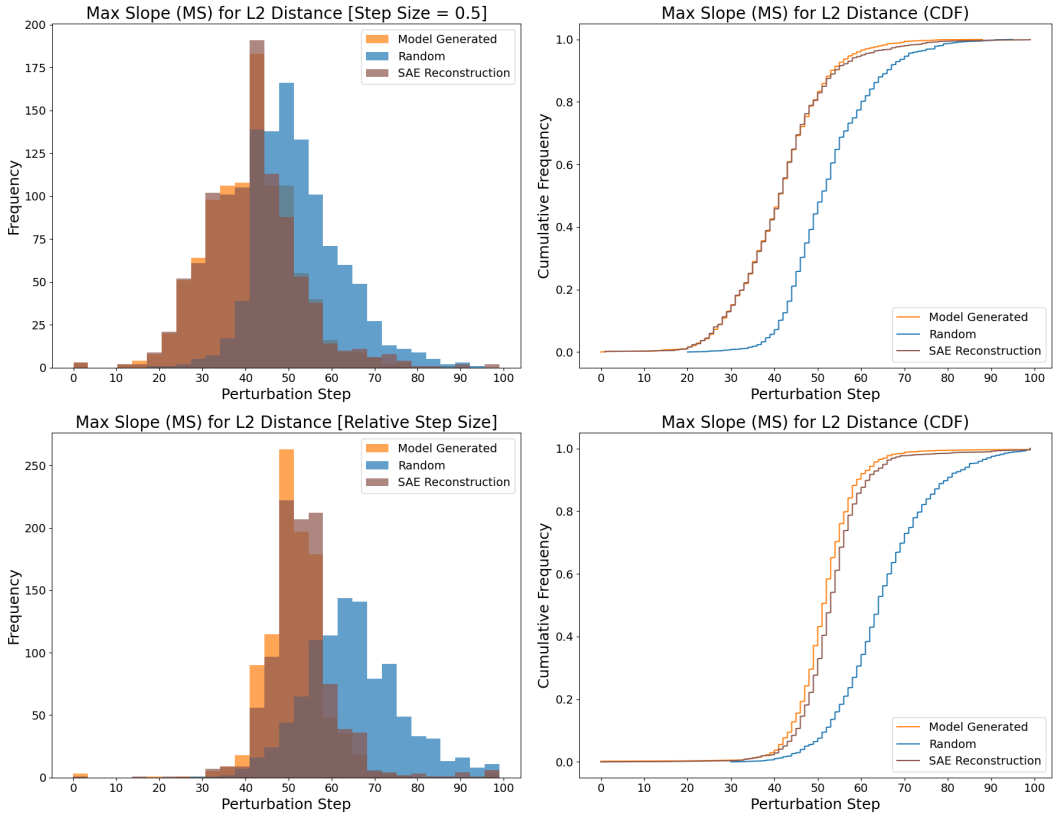

Figure D.1: The distributions of the MS steps for perturbations with absolute step size (top) and relative step size (bottom) towards random activations (blue), model-generated activations (orange), and their SAE reconstructions (brown). The left column shows the counts of MS steps occurring in different bins along the length of the perturbation, and the right column shows the cumulative frequency for the same. We find that perturbations towards model-generated activations and perturbations towards their SAE reconstructions are almost identical.

E Properties of SAE latents in model activations

We observe that model-generated activations with a low SAE reconstruction error contain approximately 21 active SAE latents on average (Figure E.1 left). The distribution is narrow around the mean and falls off very rapidly. The top latent represents around 49% of the total latent activation norm average (Figure E.1 right). The norm falls off rapidly thereafter, with the second top latent representing only around 10% on average. The distribution flattens out afterwards where latter ranks have similar contribution to the norm.

Additionally, we find that model-generated activations are made up of SAE latents that have cosine similarity to one another of approximately 0.29 on average (Figure E.2 left), with a distinct peak at 0. SAE latents primarily have positive cosine similarity to the top SAE latent, with mean cosine similarity of 0.18 (Figure E.2 right) and with a more pronounced peak at 0.

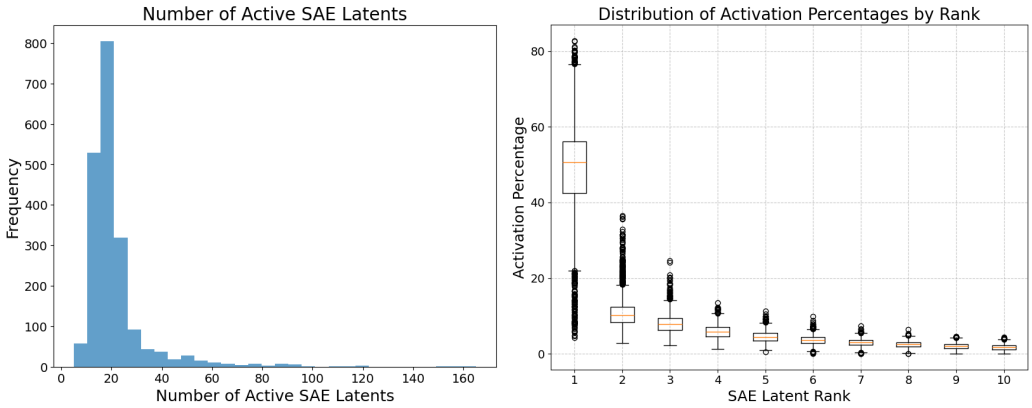

Figure E.1: The distribution of the total number of active SAE latents per activation (left) and the distribution of the percentage of the latent activation norm represented by the top 10 active latents (right) aggregated over 2000 activations.

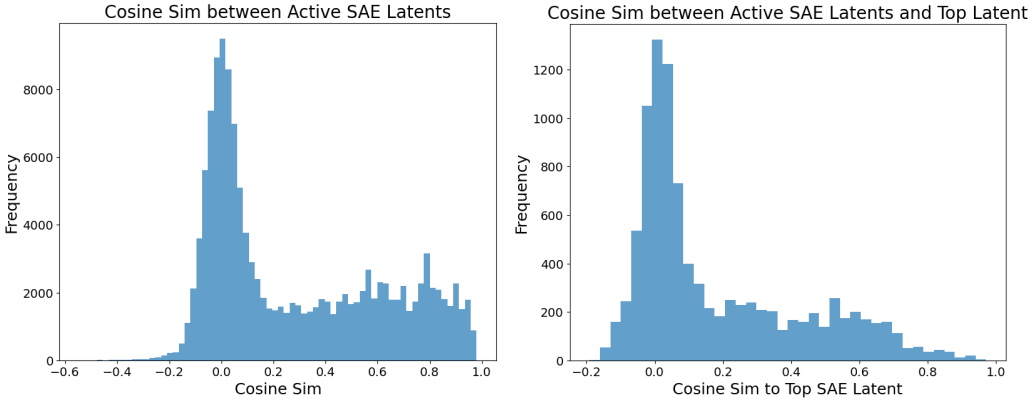

Figure E.2: The distribution of cosine similarities between all active SAE latents per activation (left) and distribution of cosine similarities that active SAE latents have with the top SAE latent (right) aggregated over 2000 activations.